4.5 Dilations

Name _____ Period ____ Date _____

Geometry 4.5 · Student sheet

I can…

- find the scale factor of a dilation and tell whether it is an enlargement or a reduction.
- dilate a figure in the coordinate plane.
- use a negative scale factor.

Vocabulary · fill in the missing word

Dilation — a _____ in which a figure is enlarged or reduced with respect to a fixed point, the center of dilation

Scale factor — the ratio of a length in the image to the _____ length in the preimage

Enlargement — a _____ with a scale factor greater than 1

Reduction — a _____ with a scale factor between 0 and 1

Part 1 · Discovery: multiply the coordinates

Plot △ABC with A(2, 1), B(4, 1), C(2, 3). Then plot the images with every coordinate multiplied by 2 and by ½.

	A	B	C	Length of $\overline{AB}$
Original	(2, 1)	(4, 1)	(2, 3)	
× 2				
× ½				

What do you notice about the sides? The angles?

Part 2 · Notes and examples (fill in each note, then work the example)

Scale factor: k = (image length) ÷ (preimage length). If k > 1 the dilation is an _____; if 0 < k < 1 it is a _____.

Example 1

Find the scale factor. Is it an enlargement or a reduction?

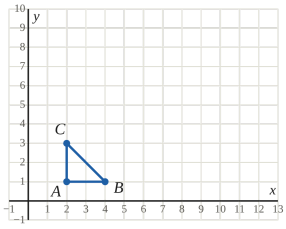
a. CP = 6 and CP′ = 15

b. CP = 12 and CP′ = 4

Dilation centered at the origin: (x, y) → _____

Example 2

Dilate △ABC with A(2, 1), B(4, 1), C(2, 3) by a scale factor of 3, centered at the origin.



Example 3

Dilate $(-4, 6)$ and $(8, -2)$ by a scale factor of $\frac{1}{2}$, centered at the origin.

Negative scale factor: the image is on the _____ side of the center (a dilation combined with a 180° rotation).

Example 4

Dilate $(1, -3)$ by a scale factor of -2, centered at the origin.

Example 5

A magnifying glass shows an insect 5 times its actual size. The insect is 0.4 cm long. How long is its image?

Part 3 · Partner practice

Partner 1

Dilate $(3, -5)$ by a scale factor of 4, centered at the origin.

Partner 2

A preimage side is 10 and the matching image side is 4. Find the scale factor. Enlargement or reduction?